

CO1 ASSESSMENT TOOL 1: CASE STUDY ANALYSIS

Smart High-Speed Rail Passenger Experience Platform

Name: TARUNIKA S

Reg.No: 192511024

Slot: D

Course Code: CSA 1007

1. Problem Understanding

Problem: High-speed rail operators must coordinate large volumes of passengers, complex train schedules and onboard facilities across an extensive network. Manual monitoring and disconnected systems lead to delayed disruption response, inconsistent passenger information, underused onboard amenities and slow incident reporting.

Introduction:

- High-speed rail is a critical mode of sustainable mass transportation, connecting cities quickly while reducing road and air traffic congestion.
- However, existing rail systems face challenges including unpredictable service delays, fragmented passenger communication, limited real-time visibility of onboard facilities, and slow response to operational disruptions.
- These challenges reduce passenger satisfaction and place a heavy operational burden on rail staff and administrators.



Objectives:

- Monitor train operations.
- Predict service delays.
- Manage passenger services.
- Track onboard facilities.
- Generate performance reports.
- Notify passengers.

2. Stakeholder Analysis

Stakeholders: Passengers, Train Operators/Crew, Station Staff, Railway Administration, Maintenance Engineers, System Administrators, Government/Regulatory Bodies.

Roles and challenges are explained through operations monitoring, delay prediction, passenger service management, facility maintenance, regulatory compliance and quality assurance.

3. Current System Analysis

Existing systems rely on manual announcements, fixed schedules and disconnected passenger information boards.

Strengths: simple to operate, low technical overhead.

Weaknesses: delayed passenger notifications, poor real-time visibility of train status, inefficient onboard facility management and no predictive analytics for service disruptions.

4. Proposed Software Solution

Modules:

- Real-Time Train Operations Monitoring
- Predictive Delay and Service Disruption Management
- Passenger Service and Ticketing Management
- Onboard Facility Tracking
- Performance Reporting and Analytics
- Passenger Notification System

5. Requirements

Functional: User login/authentication, monitor train operations, predict service delays, manage passenger service requests, track onboard facilities, generate performance reports, send passenger notifications.

Non-functional: Security, Reliability, Scalability, Maintainability, Performance, Usability.

6. Software Engineering Principles

Modularity, Scalability, Maintainability, Reliability, Security and Usability are applied using independent modules, cloud-based architecture, secure authentication, redundant backups and responsive dashboards for staff and passengers.

7. SDLC Justification

Agile Scrum is selected because passenger expectations, operational requirements and rail regulations evolve continuously. Short sprints, continuous testing, stakeholder feedback and incremental delivery reduce project risk and allow the platform to adapt to changing operational needs.

8. Expected Outcomes

- Improved passenger experience.
- Reduced service delays.
- Better operational efficiency.
- Enhanced service quality.
- Increased customer satisfaction.

9. Risks and Limitations

- Sensor and equipment failures

- Network and connectivity dependency
- High implementation and infrastructure cost
- Cybersecurity risks to passenger and operational data
- Integration challenges with legacy rail systems

10. Future Enhancements

AI-driven predictive maintenance, IoT-based seat and facility occupancy sensors, blockchain-based ticketing, multilingual voice assistants and integration with smart city transport networks.

Conclusion

The Smart High-Speed Rail Passenger Experience Platform combines IoT, AI and cloud computing to improve passenger experience, reduce service delays and enhance operational efficiency while following Software Engineering principles.

Code of Conduct:

I, [TARUNIKA S] ([192511024]), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: [TARUNIKA S]